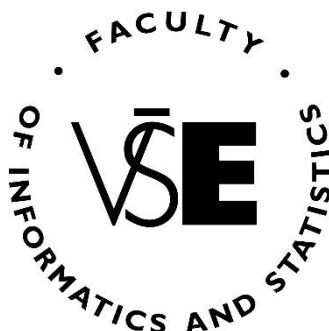


Prague University of Economics and Business

Faculty of Informatics and Statistics



**PROTOTYPE DESIGN FOR THE SETKÁVAČ
MOBILE APPLICATION ACCORDING TO THE
UI/UX DESIGN PRINCIPLES**

BACHELOR THESIS

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Abstract

This work is devoted to designing a user-friendly mobile application ‘Setkávač’ that plans in-person meet-ups. The study's first section examines the fundamental ideas of UI/UX design theory, placing special emphasis on user-centred design. The theoretical framework offers a thorough guidance for design thinking process of this application. Based on the design principles of UX and UI, the prototype of the mobile application is created during the practical stage of this work. The main aim was to design an interactive prototype in Figma program that meets the application’s purposes and user requirements. User testing and feedback loops are essential components of development cycle that guarantee that the prototype not only complies with design principles but also reflects end users' expectations and preferences.

Keywords

user experience, user interface, design thinking, prototype

Abstrakt

Tato práce se věnuje návrhu uživatelsky přívětivé mobilní aplikace „Setkávač“, která plánuje osobní setkání. První část studie zkoumá základní myšlenky teorie designu UI/UX se zvláštním důrazem na design zaměřený na uživatele. Teoretický rámec nabízí důkladný návod pro proces návrhového myšlení této aplikace. Na základě návrhových principů UX a UI je v praktické fázi této práce vytvořen prototyp mobilní aplikace. Hlavním cílem bylo navrhnout interaktivní prototyp v programu Figma, který splňuje účely aplikace a požadavky uživatele. Uživatelské testování a zpětnovazební smyčky jsou základními součástmi vývojového cyklu, které zaručují, že prototyp nejen vyhovuje konstrukčním principům, ale také odráží očekávání a preference koncových uživatelů.

Klíčová slova

user experience, user interface, design thinking, prototype

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List of Abbreviations

UI User Interface

UX User Experience

Introduction

The pervasiveness of digital technology in the modern days has completely changed how people engage with the environment, making it more difficult to separate between the real and the virtual. A notable example of this technological advancement is the widespread shift towards an online-centric life. Almost every component of an everyday routine of many people, from work to grocery shopping, has shifted to being online. The field of User Interface (UI) and User Experience (UX) design is of utmost importance in the quickly changing digital landscape.

The motivation behind choosing to design the application ‘Setkávač’ dedicated to planning meetups stems from a personal experience of a foreigner navigating the social dynamics in a new country and is rooted in desire to create a platform that offers an opportunity for bringing people together, thus facilitating cultural and language exchange or, more importantly, making personal connections more accessible.

Another motivation comes from the desire to improve my understanding and practical proficiency in UI/UX design. This bachelor's thesis is essentially the result of a convergence of personal narrative and an objective for design related career.

Goal

The purpose of a bachelor's thesis devoted to UI/UX prototype development for a mobile application 'Setkávač' is to:

- Understand the general approaches for developing mobile applications
- Explore the fundamental ideas of UI and UX and gain an understanding of the theoretical foundations that guide the creation of efficient interfaces.
- Choose the best UX method and create a UI/UX prototype following the approach
- Test the prototype, gather the results of the testing and define improvement points for the prototype

Methodology

Chapter 1 of this bachelor's work takes a general look at the mobile applications, describing different ways how to implement mobile applications and comparing them. First part of Chapter 1 leads us to the conclusion why the strategy to develop a mobile application for IOS has been chosen. Also, in this chapter we define UI and UX and other definitions that will be used later on.

In chapter number 2, we analyse different UX Methodologies, such as Design cycle, Design sprint, Double diamond and Desing thinking, choosing the necessary methodology for our practical part. Also, there is a description of Heuristic principles that will be used in further steps of this bachelor's work.

In chapter number 3, called Design process, we are realising the practical part of this work, starting with Empathy stage up to Testing stage using different approaches like competitor analyse, user research, creating of user journey maps and applying the concept of "How might we". After it, we ideate the potential improvements of a future prototype, create this prototype and test it. Discuss some points for a future improvement.

The bachelor's work finishes with a conclusion.

Research

This bachelor work is inspired by many resources including:

Steve Krug – Don't make me think

Steve Krug's user experience and online usability manual (Krug, 2014) puts a strong emphasis on simplicity and user-friendly design concepts. He believes that navigation should be intuitive, and navigation systems straightforward and clear. This book is an ultimate guide for websites that prioritize a user-centric approach.

Don A. Norman – The Design of Everyday Things

Don Norman's "The Design of Everyday Things" (Norman 2013), is a classic book on the subject of human-centred design. Norman's book delves into the design principles that enable people to find common products interesting and useful. He presents the idea of "user-centred design" and highlights how crucial it is to develop products that complement how people naturally view and interact with the environment.

Jeffrey Rubin, Dana Chisnell – Handbook of Usability Testing

The book offers a thorough guide for planning and carrying out efficient usability tests, covering the concepts and procedures of usability testing. It explores all facets of usability testing, from organising to participant recruitment to test scenario creation, session moderating, result analysis, and reporting.

Redesign a návrh prototypu mobilní aplikace – Adam Šindler

The purpose of this bachelor thesis was to develop a graphic user interface for application Kimobil, which was a way to improve the old version. For this purpose he has selected the Design thinking approach and used a number of various methods during different stages of development.

1 Definition of mobile application

A mobile application is a type of software designed to run on a mobile phone, such as a smartphone or tablet PC. Applications are typically small, standalone software units with limited capabilities. This use of application programming was originally promoted by Apple and its product called *App Store*, which offers a huge number of applications. (Bartosińska, 2024)

Unlike desktop applications, mobile applications are separated from in-built software systems. Apart of other factors, each universal application provides a distinct and limited utility. For example, it is usually a game, a number calculator, or a portable Internet browser. (Bartosińska, 2024)

Early mobile application provided general information and information services on the global web, which included email, calendar, stock market, weather information. However, the demand of mobile users, along with the possibility of developing a mobile application, extends to other categories, such as mobile games, production automation, GPS and others. The rapid growth in the number and variety of fields of application has led to the emergence of large and diverse directions. Many services today require the help of mobile application technologies, such as location, online banking, tracking, ticketing and even mobile health services. (Wikipedia contributors, 2024)

The simplest mobile apps take PC-based apps and bring them to a mobile device. As mobile applications become more and more popular, this strategy becomes quite imperfect. A more modern methodology provides clear growth for the mobile environment, taking advantage of its limitations. For example, applications in which the main functional areas seek to work from the earliest stage with a focus on mobile devices, taking into account that the client is not tied to any area, as when using a PC. (Zigmos, 2015)

1.1 Types of mobile applications

1.1.1 Native applications

A native mobile application is one type of application in which it is created and developed for a specific type of device platform, such as Android or IOS (Figure 1), using a specialized coding language. To create a native app, the developers' chosen coding language must have access to the device platform. Typical app features for this category might be offline mobile games, dictionary apps, etc. (Lee et al., 2010)



Figure 1 - Native application

The main advantage of native applications is a high-quality user interface. Considering all of this, the designers who create them use their own UI devices. Access to a wide range of APIs also helps speed up development and expand the use of applications. Native apps must be downloaded from app stores and directly embedded into devices. That is why they must go through a restricted review process before publishing. (Zammetti, 2011)

The application usually downloads all the images so that the user can use the application without an internet connection (some apps require Internet connection to proceed some functionalities). Another common example of native mobile apps that everyone knows is Facebook. (Iversen & Eierman, 2014)

1.1.2 Web applications

Web apps are software applications that work together with their own mobile apps and run on mobile devices. However, there are significant differences between native and web applications. Web applications use browsers to run and are usually written in CSS, HTML, JavaScript. (Grigsby, 2018)

Such apps redirect the customer to a URL and then ask them to select an app. As a result, web applications force customers to bookmark such a page for future viewing. That is the reason why they require less amount of memory. (Grigsby, 2018)

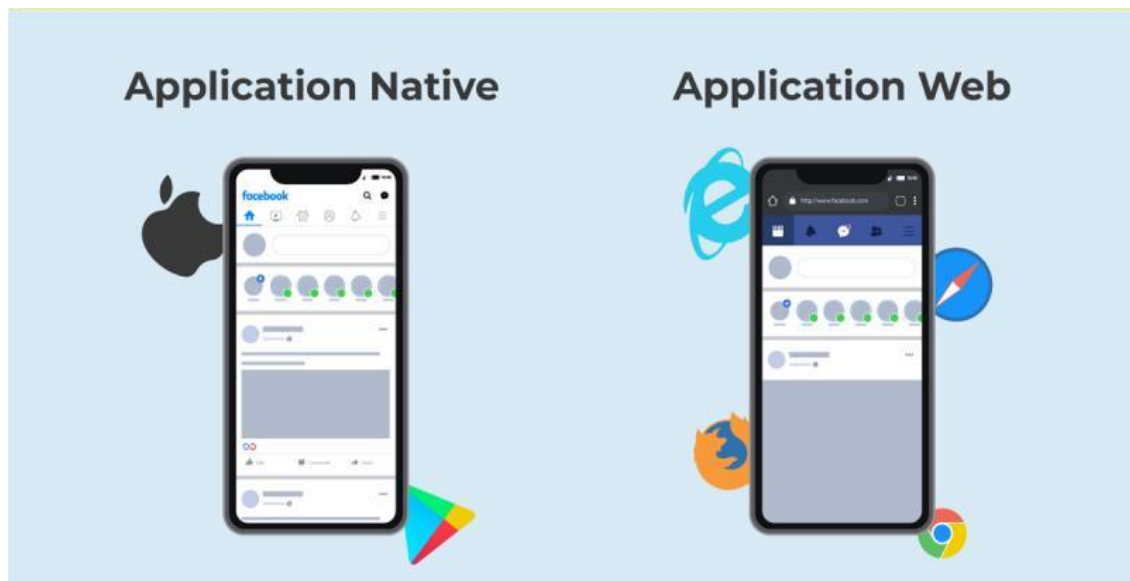


Figure 2 - Native and Web applications

Web apps use the same organization method as native apps but are accessed through a website browser on a mobile device (Figure 2). They are not stand-alone programs in terms of downloading and installing code on the device. These are truly responsive websites that adjust their user interface to the client's device. When you choose to "install" a web app, it often bookmarks the site's URL on the device. (Firtman, 2015)

1.1.3 Hybrid applications

A hybrid app combines the benefits of a mobile web app and a native app. It is built using HTML, CSS, Javascript, works on mobile WebView. However, the hybrid app can still use the device's features like capture, GPS, vibration, etc. (Griffith, 2017)

These are web apps that closely resemble native apps. They may have an app icon on the home screen, a responsive design, be fast, even have the ability to work offline, but they're actually web apps designed to look native.

1.1.4 Difference between Native and Web applications

Native and hybrid apps are two types of mobile apps. Despite the external similarity and the design behind them, the technologies are completely different. As the name implies, hybrid apps combine online applications with native mobile apps. HTML, CSS, and JavaScript are web technologies that can be used to create them. They are distributed through app stores, where consumers can download them as native apps for Android or iOS. (Chandran, 2024)

The main difference between hybrid and native applications is that hybrid applications are built on all platforms, while native applications are developed for specific operating system

(Figure 3). Unlike a unique app for each smartphone operating system, a hybrid app is the same for all platforms and works equally well on each of them. (Chandran, 2024)

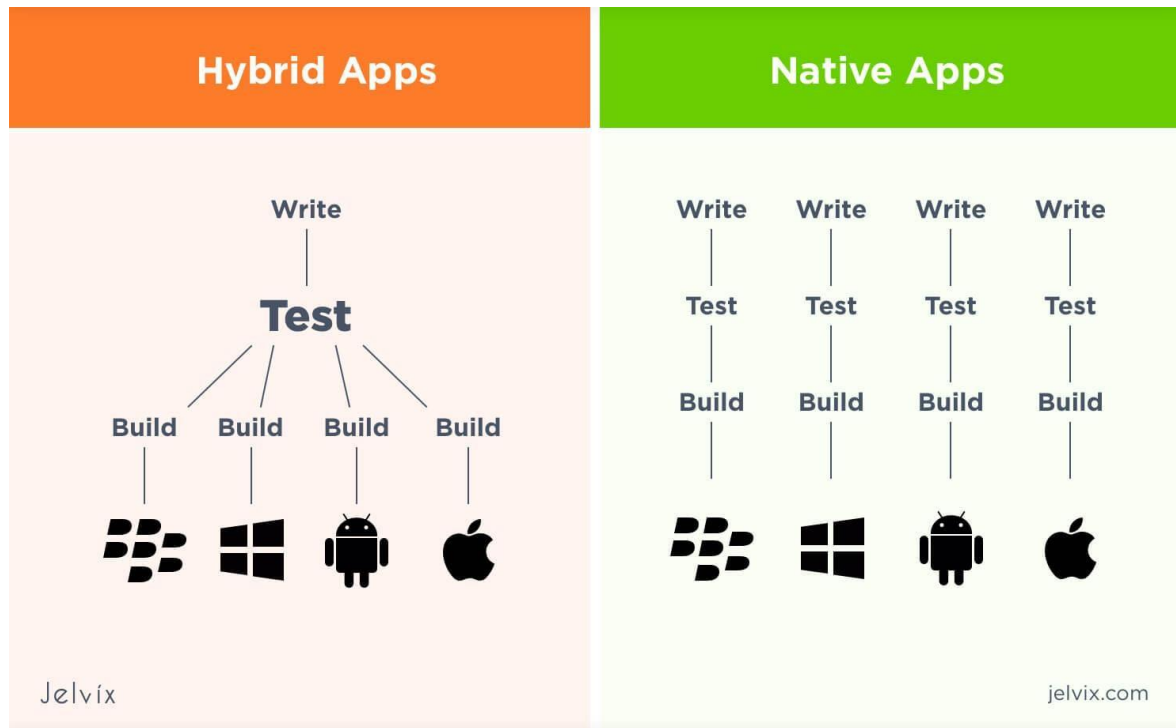


Figure 3 - Difference between Hybrid and Native applications

Advantages of hybrid applications include portability and ease of development, you only need to write the code once, and hybrid software will run on many operating systems. For cross-platform application development, you can choose hybrid platforms such as Ionic or Cordova. As for native mobile apps, they should be built in platform-appropriate languages such as Java, Swift, or Objective-C. (Griffith, 2017)

In addition, native mobile apps can access the smartphone's built-in features, such as the camera and microphone. A hybrid app relies on plugins like Cordova to take advantage of the device's built-in capabilities.

Here are the main benefits of using Native application approach:

- **Unified UI/UX.** With the help of native applications, users can shorten and simplify the process of getting used to the program, because in general UI will be standard. Designers and developers also find it easier to apply the best standards and practices compared to hybrid applications. (Meier, 2018)
- **Access to the full range of device functions.** Since each native app is platform-specific, the app user can directly access the device's hardware like microphone, camera, GPS, etc. Also, native apps have a big advantage in push notifications that come through APNS (iOS Server) that require an app bundle ID, and similarly GCM (Google Cloud Messaging). (Meier, 2018)
- **Scaling.** With high compatibility with a single platform, native applications can be quickly configured, making them easy to scale. (Meier, 2018)

- **Autonomous performance.** Native applications can work effectively without an Internet connection, because of the fact that content of the program is downloaded directly during installation and operation. Additionally, these apps have robust in-browser caching features that make all cached resources available offline. (Meier, 2018)

1.2 UI/UX Design

The appearance and usability of a mobile application is one of the main priorities for users. And that's when the UI/UX design of the application comes forward. Steve Krug (2014) believed that a good design should be intuitive, and allow users to navigate without difficulties or effort that is unnecessary. Before moving on to the app design block, it is necessary to first understand two important design terms, UI and UX.

1.2.1 Historical Context of Human-Computer Interaction

When computing first began, most user interaction was text-based and took the form of command-line interfaces (CLI), which required users to memorize and become familiar with complex syntax in order to perform tasks. For seasoned users, the CLI was effective, but it presented a major obstacle to entry for newcomers. However, in the 1970s and 1980s, pioneers like Xerox PARC and Apple, with its Lisa and Macintosh, started the GUI revolution, which introduced a paradigm shift by using windows and icons to represent information and actions graphically. (Diaz, 2023).

A psychological perspective of design that first appeared in the early 20th century is Gestalt psychology.

1.2.2 Gestalt psychology

Gestalt psychology stressed the idea that people's perceptions and understanding of the world are meaningful wholes rather than the sum of their individual sensory experiences. Pioneers of Gestalt psychology included Max Wertheimer, Kurt Koffka, and Wolfgang Köhler held the view that people organize stimuli into patterns and structures in order to perceive and interpret them rather than seeing them as (Yablonski, 2024)

Common region

Objects located within the same area, are perceived as a group. The common region principle is particularly useful in interface design: it helps to group information and organize content, but it also allows to separate one group of content from another or create a focal point.

The general area reinforces the hierarchy, increases the visual organisation of the content and helps to find the information you need. (*What Are the Gestalt Principles?*, 2024)

Proximity

Elements located closer to each other are perceived more connected than those at a long distance. That is why we often perceive individual elements as a group, rather than individually. We use the principle of proximity in UI design to organize content, group related information, or arrange elements on the screen. (*What Are the Gestalt Principles?*, 2024)

The principle of proximity says that interconnected objects should be located close to each other, and unrelated objects should be at a noticeable distance. In this case, white space plays a very important role, creating a contrast that directs the user's gaze in the right direction. It greatly helps to set the direction of attention and reinforces the visual hierarchy. As a result, such a screen is easier to read and scan - and this helps users to achieve their goals faster and immerse themselves more deeply in the content of the interface. (*What Are the Gestalt Principles?*, 2024)

Closure

We often perceive a group of elements as one familiar object or a figure. The principle of closure also works when the object is incomplete, or some parts are connected to each other. The essence of the principle of closure is that when the necessary amount of information accumulates in the brain, it immediately comes to conclusions, that is, we fill in the gaps and create a general image of the object in our heads. In this way, it is possible to convey the same information using fewer elements and at the same time the interface will become less complex and overwhelmed. (*What Are the Gestalt Principles?*, 2024)

Similarity

Elements that have similar visual characteristics seem more related to a person than those that look different. We tend to think of similar elements as a group or pattern. We even assume that such elements have the same purpose.

The principle of similarity helps to classify and organize objects within a group and link them to one another by function or meaning. (*What Are the Gestalt Principles?*, 2024)

Continuity

Elements built along one straight or smoothly curved line seem to human eye more interconnected than those located randomly or along a broken line.

Elements located along a line are perceived as grouped. The smoother the line, the easier the elements are defined as a single figure, because our brain likes to follow the path of least resistance. (*What Are the Gestalt Principles?*, 2024)

1.2.3 User Interface (UI)

A user interface (UI) is the point at which human users interact with a computer, website, or application. The goal of an effective user interface is to make the user experience simple and intuitive (Hannah, 2023)

The user interface is created at levels of interaction that appeal to human senses (sight, touch, hearing, etc.). These include input devices such as keyboard, mouse, trackpad, microphone, touchscreen, fingerprint scanner, electronic pen, and camera, and output devices such as monitors, speakers, and printers. Devices that interact with multiple senses are called "multimedia interfaces." For example, the everyday user interface uses a combination of tactile input (keyboard and mouse) and visual and audio output (monitor and speakers) (Dash, 2022)

1.2.4 User Experience (UX)

Alongside the growth of the World Wide Web and the need for improved interaction design, the field of human-computer interaction (HCI) gave rise to user experience (UX) design. A pivotal figure in the early development of UX design is cognitive scientist and psychologist Donald Norman (Yablonski, 2024).

While at Apple, Norman stressed the significance of designing products that are not only functional but also provide the user a positive and satisfying experience. He would later coin the term "user experience" to encompass all aspects of the end user's interaction with a system, including the interface, graphics, industrial design, physical interaction, and more (Yablonski, 2024).

The UX design of an application is the way the user interacts with the elements of the application interface. User Experience (UX) design combines logic and critical analysis to understand and plan user's interaction with products, systems, or services. UX design is comparable with the plan of a house. It cares about how one comes in and comes out of the house and one's route from one room to another. The main goal of a UX designer is to connect all the elements of the user interface and provide a user-friendly navigation. The only thing a UX designer focuses on is a holistic understanding of how the user wants to interact with the application.

The difference between UI and UX is quite simple: UI designers work to define how the user interface will look, while UX designers take responsibility for how the user interface works (Figure 4).



Figure 4 - Difference between UI and UX

2 UX Methodology

The field of UX design employs a variety of methodologies of design development to help designers better understand user needs and create digital products that meet those needs. A few of these methodologies are covered in the following chapter.

2.1 Design cycle

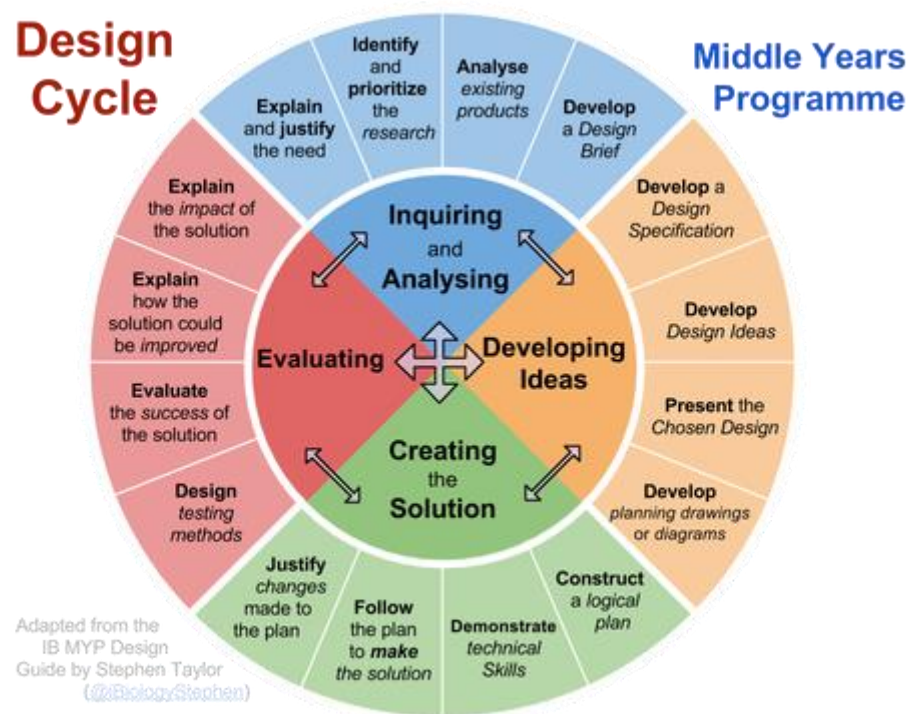


Figure 5 - design cycle

The Design Cycle is a methodical approach that takes an idea from inception to completion and focuses on feedback and continuous improvement (Boicheva, 2023).

Plan

The first part of design cycle includes research, analysis, and planning, where the design goals, objectives, a strategy for solving problems, an awareness of the target market and competitors are determined (Boicheva, 2023).

Develop

The many ways to develop design ideas: brainstorming, analysing competitors' work, making storyboards, coming up with innovative designs, and more. After the concept is decided, it is necessary to refine it and provide additional context through storyboards,

labelled diagrams, blueprints, or any other visual aid that might help to visualize it clearly (Klemen, 2020).

Create

This stage begins with constructing a plan which will be then followed. The main output of this stage is creating a prototype that will be evaluated during the next stage.

Evaluate

This stage evaluates created prototypes during usability testing in order to find out of the design is easy and intuitive to use with full documentation of the testing process.

2.2 Design sprint

A design sprint is a process that lasts 5 days and is about new ideas and problem-solving challenges in a collaborative manner.

Day 1: Map

In the first day, designer team and other participants will uncover why are they solving this problem, how to get everyone on a same page with what they're trying to achieve, map out customer experience for current product or service, map data that was used

Day 2: Sketch

On day two they would generate solutions in an attempt to get a better understanding of the problem. They will be inspired and create a number of different solutions to the original problem. Everyone on the team will sketch his or her solution individually. (Staff, 2024)

Day 3: Decide

From all of the ideas generated on Day 2 they need to decide for one. To do this, a five-step process called 'Sticky decisions' is used.

Day 4: Prototype

Day four is set aside to build a prototype of their solution so that they can see if it will work before wasting any additional money, time, and resources. The solution that they select will require a prototype for someone to use. (Staff, 2024)

Day 5: Test

They test the prototype on 5 real users who evaluate if the solution is worth putting more time into it.

2.3 Design thinking

2.3.1 Empathise

Empathy is essential to design thinking because it allows designers to set aside your assumptions about the world and gain insight from the users and their needs. The team's goal is to understand the problem, usually through user research (What Is Design Thinking?, 2024).

2.3.2 Define

After gathering data, they examine the observations and synthesize them to define the core problems, which are called problem statements. The team may develop personas to stay focused on human-centred approach (What Is Design Thinking?, 2024).

2.3.3 Ideate

Once the foundation is in place, teams get ready to "think outside the box." They come up with creative solutions to the problem statement and different perspectives on the issue (What Is Design Thinking?, 2024).

2.3.4 Prototype

This is an experimental phase when the team creates simplified versions of the product (or specific features of the product) to test out ideas; these could be as simple as paper prototypes. The goal is to find the best possible solution for each problem (What Is Design Thinking?, 2024).

2.3.5 Test

The prototypes are tested with actual users to see if they fulfil the requirements; if not, the team changes the prototype or even goes back to the Define stage to take a different look to the problem based on new insights from the test (What Is Design Thinking?, 2024).

2.3.6 Conclusion

There are many interpretations of Design thinking, but in this work, I will particularly focus on the approach of the Hasso-Plattner Institute of Design at Stanford, which is also known as d.school. I have chosen this method to follow along while developing the application in the practical part of this work in chapter 3.

2.4 Heuristics

Heuristics are 10 general principles for creating a user interface that were offered by Jakob Nielsen firstly in 1994. Even though the article was edited in 2020; the main 10 principles stayed the same. In the list below you can see these 10 principles with an explanation. (Nielsen, 2024)

- **Visibility of System Status**

Users should always be informed about current state of the application by the design, with suitable feedback provided in a short time (Nielsen, 2024).

Example:

In shopping malls, info tables usually have a sign where are you currently located (Nielsen, 2024).



Figure 6 - Info board in the shopping mall

- **Match Between the System and the Real World**

Instead of using internal jargon, the design should speak the language of the user. It should try to be as much as possible to real-world conventions, arranging information in a logical and natural order, and use words, phrases, and concepts that the user is familiar with (Nielsen, 2024).

Example:

Users can quickly figure out which stovetop control corresponds to which heating element when the layout of the controls matches the layout of heating elements (Nielsen, 2024).



Figure 7 - stovetops layout example

- **User Control and Freedom**

Because users make mistakes, it is important to provide a clearly marked "emergency exit" so that users can quickly stop an unwanted action without having to go through a long process (Nielsen, 2024).

Example:

Emergency exit door with a recognizable label in all buildings (Nielsen, 2024).

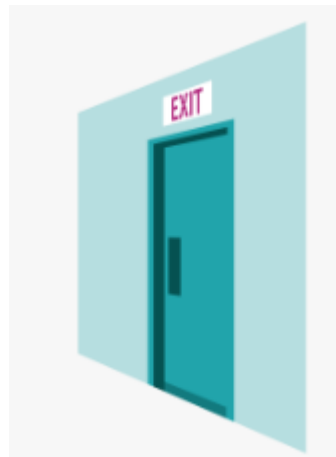


Figure 8 - Emergency exit door

- **Consistency and Standards**

Stick to platform and industry conventions so that users don't have to wonder if various words, situations, or actions mean the same thing (Nielsen, 2024).

Example:

Check in tables are usually located at the front of hotels – it is something that all the visitors expect to see (Nielsen, 2024).

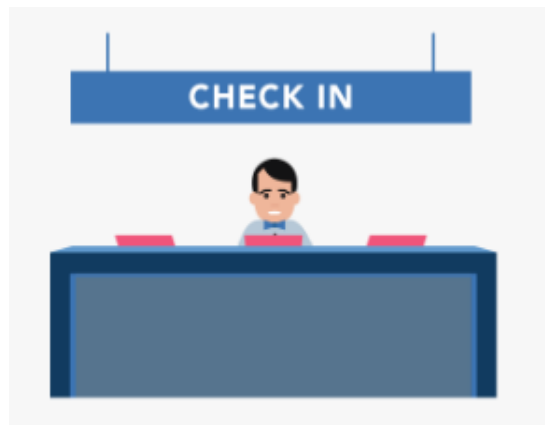


Figure 9 - Check in at the hotel

- **Error Prevention**

While well-crafted error messages are crucial, the best designs take extra care to avoid issues in the first place; they either remove error-prone conditions or look for them and give users a confirmation option before committing to an action (Nielsen, 2024).

Example:

On winding mountain roads, guard rails keep cars from falling off cliffs (Nielsen, 2024).

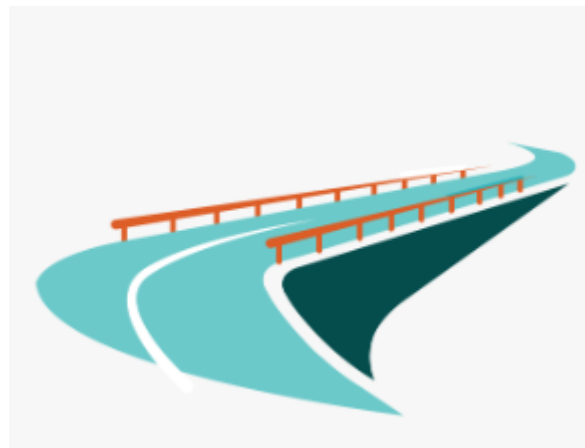


Figure 10- Guard rails on the road

- **Recognition Rather than Recall**

Reduce the amount of information that the user must manage by making elements, actions, and options visible. Information that is necessary to operate the design (such as menu items

or field labels) should be available or visible when needed. The user shouldn't have to memorize information from one area of the interface to another (Nielsen, 2024).

Example:

People are more likely to correctly answer the question Is Lisbon the capital of Portugal? rather than What's the capital of Portugal? Most people find it easier to recognize the capitals of countries than to remember them (Nielsen, 2024).



Figure 11 - Building that helps to recognize the place

- **Flexibility and Efficiency of Use**

Allow users to customize frequent actions; shortcuts, hidden from new users, may speed up the interaction for the more experienced users so that the design can accommodate both new and experienced users (Nielsen, 2024).

Example:

Maps show the regular routes, but locals who are familiar with the area can find shortcuts (Nielsen, 2024).



Figure 12 - Map example

- **Aesthetic and Minimalist Design**

Information that is unnecessary or infrequently needed should not be present in interfaces; each additional unit of information in an interface reduces the relative visibility of the relevant units of information and competes with them (Nielsen, 2024).

Example:

Excessive decorative elements, such as an uncomfortable handle or a difficult-to-wash nozzle, can decrease the usability of a teapot (Nielsen, 2024).



Figure 13 - two examples of teapot

- **Help Users Recognize, Diagnose, and Recover from Errors**

Error messages should accurately describe the issue and offer a workable solution. They should not use error codes (Nielsen, 2024).

Example:

Roadside wrong-way signage alerts motorists that they are going the wrong way and requests them to stop (Nielsen, 2024).



Figure 14 - wrong way sign example

- **Help and Documentation**

The system should work great without further explanation, but in order to help users understand how to accomplish their tasks, documentation might be required (Nielsen, 2024).

Example:

Airport information kiosks are easily recognizable and provide context-based solutions to customers' problems (Nielsen, 2024).

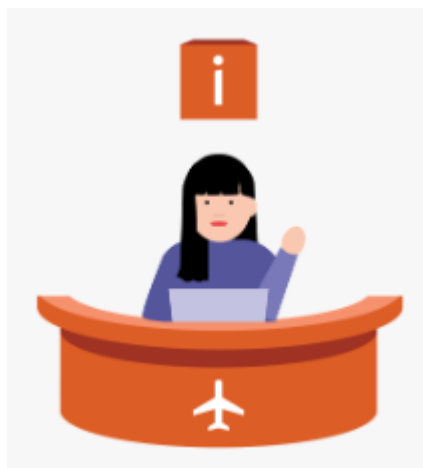


Figure 15 - Info kiosk at the airport

3 Design process

This chapter is dedicated entirely to the practical part of my work, where I will be following the Institute of Design at Stanford's approach of Design thinking process, separate stages of which were defined in chapter 2.3.

3.1 Empathise

First and the cornerstone of the whole process is Empathise, especially if the application being developed is a socially-oriented product. This stage requires a deep dive into the lives of possible target audience and try to be weigh using their glasses — analyse what people want, need, fear, etc. Tim Brown (2019), CEO of IDEO, indicated that empathy is a mental habit that leads to the insights and innovation (p. 47). This thought emphasises the essential importance of understanding users not only on the surface level, but on emotional level as well.

Empathy stage in context of the development of Setkávač aims at comprehensive research of users' needs in the area of creating or participating in social events. This is particularly relevant in light of the shift in social dynamics due to COVID-19 and surge towards the digitalisation of the world. Sociologist Sherry Turkle (2017) believes that "technology is not only changing what we do but who we are" (p. 26).

3.1.1 Competitor analysis

Through examining apps made by competitors, UX designers can find out about the features and functions that users of similar applications usually expect. This understanding is helpful in realising what users need and like, thus guiding the creation of a new app. Also, looking at rival apps assists UX designers in setting benchmarks according to industry norms and good methods. Designers, by studying the good and bad points of competitors' apps, can set a standard for how their own app should perform and be experienced. Looking at other apps might help in understanding about user interface design aspects like navigating around or arranging layouts. At the same time, thinking over successful design models can guide designers during creating process.

For competitor analysis in my UX process, I have chosen Facebook Events and Meetup.

Facebook is as a major social media website that is commonly known among almost everybody.

Also, I have selected Meetup as another competitor for the examination. Meetup is famous for concentrating on helping people arrange group meetups centred around similar interests, hobbies and activities. While Facebook Events covers a wide range of events, Meetup is known for encouraging community interaction through planned gatherings.

‘Setkávač’ has a main focus on connecting people who share similar interests as well, making Meetup the closest competitor by its functionality. Through studying Facebook Events and Meetup, I’m trying to understand the overall norms and design methods used in popular applications of such type.

Facebook Events

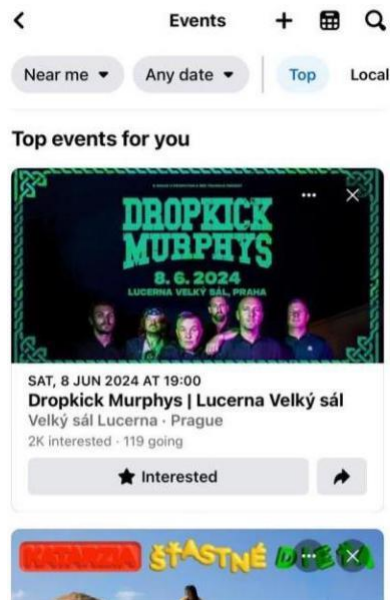


Figure 16 - facebook events

Pros: Convenient search filter, engaging – a user can see which events his friends are going to attend, a user can create both - private and public events, free promotional tool for businesses.

Cons: Some users have expressed their disappointment as the list of attendees of any event became closed for public eye several years ago, and only the organizer is now allowed to see the guest list. It is done due to safety reasons as Facebook is the biggest social media platform with almost a lifetime of personal information on display.

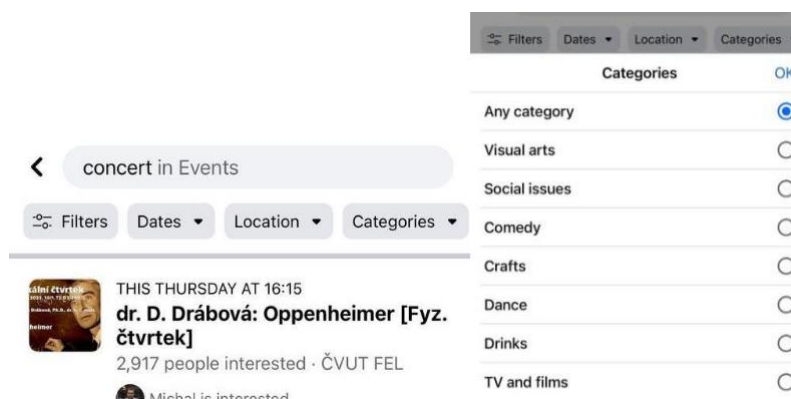


Figure 17 - search by filters in facebook

Users can not search events by categories unless they have already looked up a specific event. In my opinion, people who type in something they want to find in a search bar need

the filter by categories less than people who try to find something to their liking on the explore page with no specific event name in mind.

Meetup

Meetup is a social networking platform that allows users to discover and join the groups that align with their individual interests. Meetup is used to arrange offline, in-person gatherings in users' region, as opposed to only having online conversations about these interests.

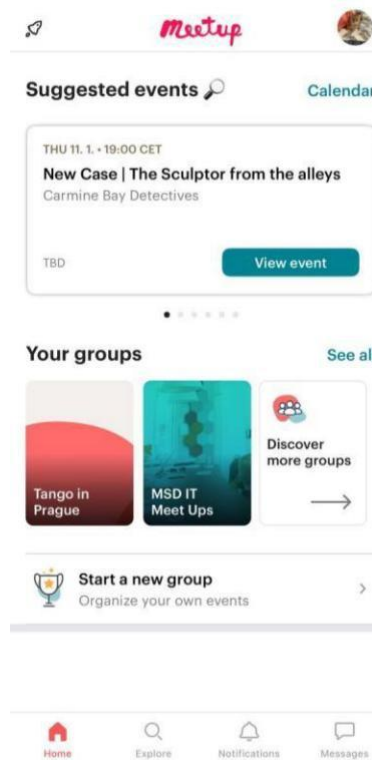


Figure 18 - meetup homepage

Pros: allows users to look for events on a map, users can see all the attended events, not only events but also groups by interests, personal profile questionnaire with ready-made answers to choose from.

Cons: The primary explore page is in the second tab, while the home page functions as a "my profile" page. Its sections include "Your groups," "Your interests," and "Your calendar." Therefore, if users are opening the app for the first time and are curious about the content they might find, they'll need to take an extra step by clicking on the second tab to the right. In order to become a host, one has to pay for a premium account, but the app does not give this important information until users have a form of an event they would like to organise already filled out, ready to publish.

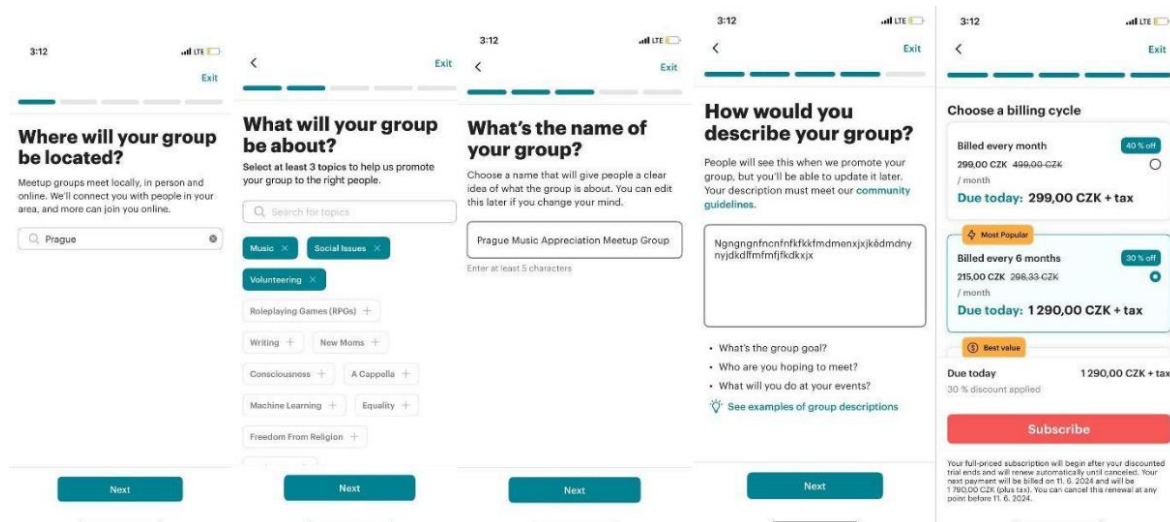


Figure 19 - meetup creation of the event

3.1.2 User research

User research is an integral step in the entire process of product development specifically for UI/UX design. The purpose of this phase is to set up the main user target beforehand, learn what are their needs, expectations, pains and motivations which will help in creating a product that meets the needs and expectations of its end users.

The method of conducting interviews with potential users is very important to collect qualitative data during the empathy phase. This approach allows to understand users' motivations, problems and needs, which cannot be obtained using quantitative research methods.

The research took the form of semi-structured interviews between the author and interviewees. Interviewees were selected from among volunteers who fell under one of the target segment groups that were defined earlier.

Semi-structured interviews are a qualitative research method used to understand human attitudes, behaviours and beliefs. Semi-structured interviews employ a set of predetermined questions matched with free-form conversation to get more detailed perspectives about the viewpoints, attitudes & experiences of informants (*What Are Semi-Structured Interviews?*, 2023).

For conducting the interview, a semi-structured guide of pre-planned questions with flexibility for exploration. The pre-planned questions included:

What experience do you have regarding social events?

How was your experience with applications of similar type?

Which features would you like to see in a perfect application for event organisation and discovery?

The semi-structured interviews were conducted with 8 interviewees that were randomly selected from volunteers who had at least some experience with similar applications and wanted to participate in the interview. Interviewees were in the age range of 18-51 years and half were men and half women.

Every interview lasted from 30 to 50 minutes which allowed to explore experience and thoughts of interviewees on the deeper level (see Appendix A).

The results revealed following insights:

- Several interviewees have some problems with searching for events that would fit to their interest. This signals the necessity of more precise event search.
- Some interviewees have mentioned that platform like Facebook Events is too integrated into social network, which scares off the users that value privacy.
- The majority of interviewees are looking for a tool to create events in quick and easy manner.
- Several interviewees shared they wanted more informal communication opportunities.
- The majority of interviewees expressed their anxiety before attending new events, especially if they don't know the other participants.

Thus, the information, obtained from the interviews and studying competitor applications, helped in understanding which segments of users are involved in similar services and get three groups of potential users of the application the target groups were defined:

Table 1 - target users

Segment	Age	Needs interests	Motivation
Students	18-24	Entertainment, educational events	Learning, satisfying curiosity
Young professionals	25-35	Networking, professional development, social interactions	Meeting new people, career development
Active middle-aged people	30-50	Event management, audience engagement, marketing	Business growth, customer attraction

Based on the data collected, I have created empathy maps for different segments of potential users of "Setkávač". An empathy map is a powerful visual tool that helps designers organise information about users and gain a deeper understanding of their needs, thoughts, and feelings. Dave Gray (2010) points out that "an empathy map helps to go beyond demographics and create a deeper, more personal understanding of the user" (p. 65).

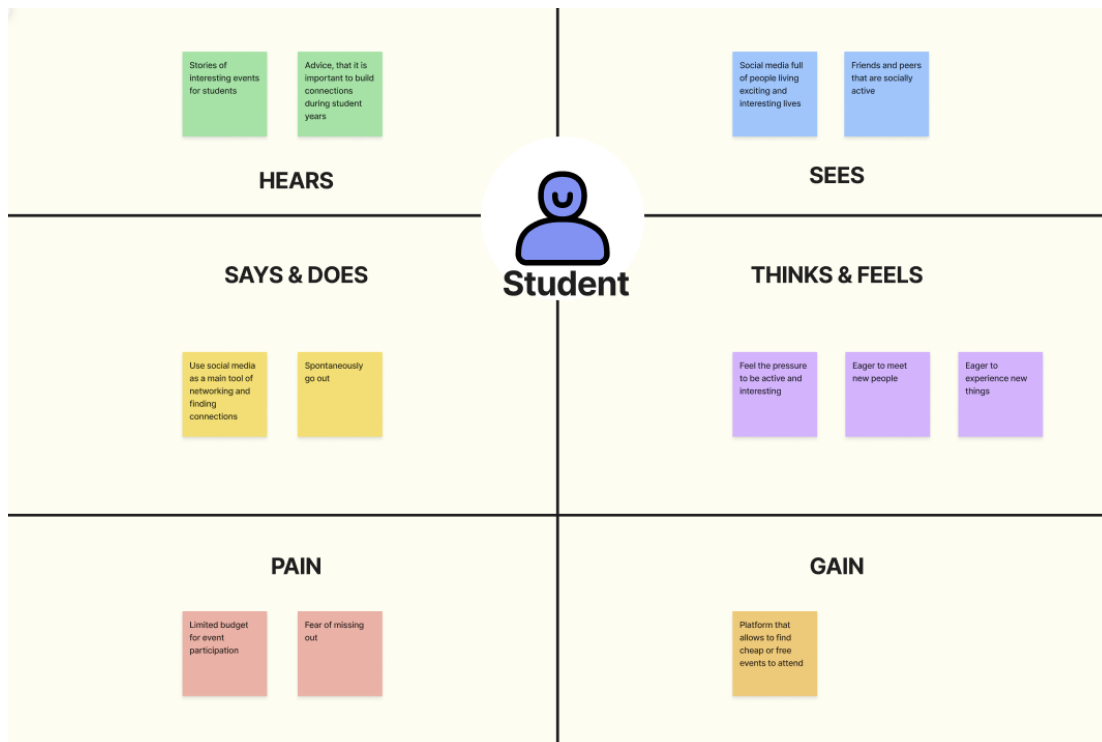


Figure 20 - empathy map student



Figure 21 21 - empathy map young professionals

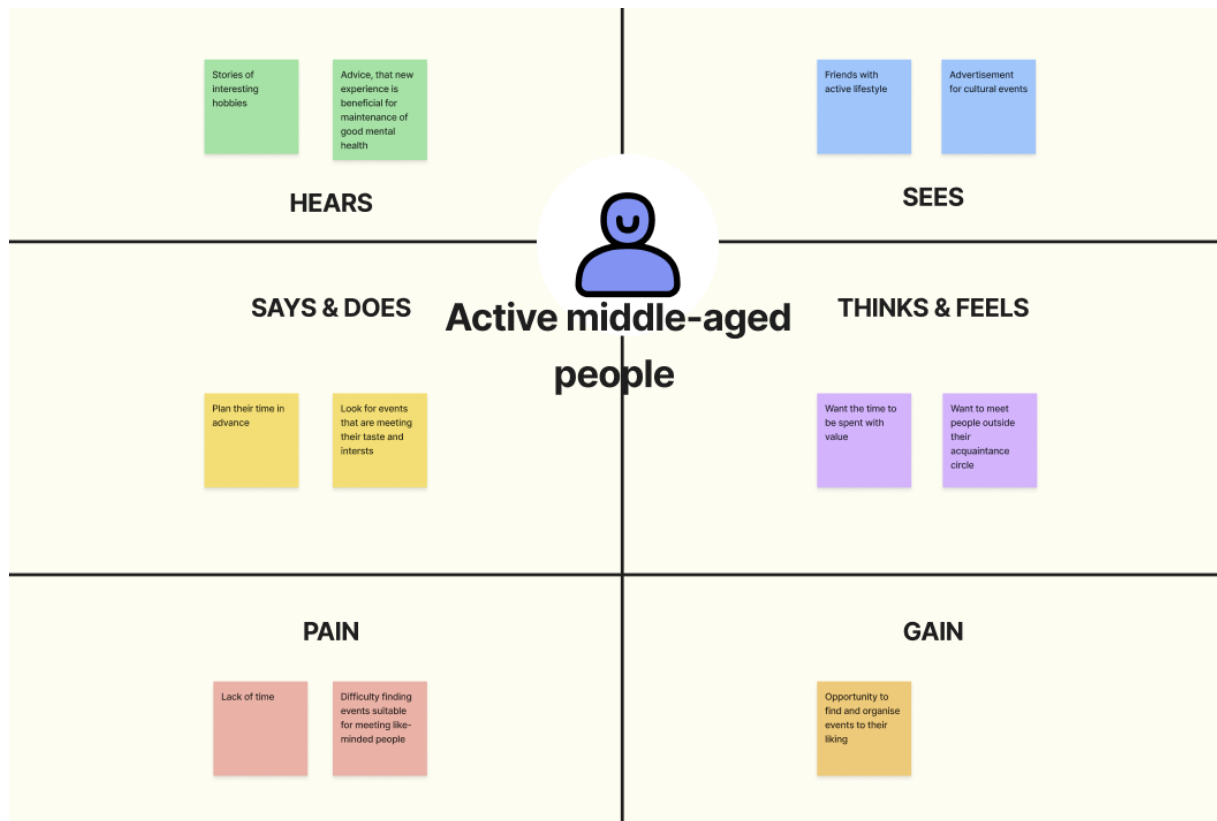


Figure 22 22 - empathy map active middle-aged people

Empathy maps have helped in defining some insights about the target audience. For instance, middle-aged people can be anxious when entering unfamiliar social environments. Such insight suggests a functionality where social interactions are possible before the event, e.g. chat rooms. Some of the students have expressed their desire to organise mutual studying sessions. All targeted groups have expressed their need to find events that would meet their field of interest or create an event of their own.

All in all, in order for any product to succeed we must know enough information about the users, which the application is intended for, and their needs — what they want, feel, hear, do, what their wishes and pains are. These insights put together create a clearer picture of what “Setkávač” should have when it comes to meeting its users’ needs.

3.2 Define

The problem definition stage is critical in the design thinking process because it transforms the insights gained in the empathy stage into well-articulated problems that need to be addressed. As John Arnold notes in Creative Engineering (2016): "A well-formulated problem is a half-solved problem" (p. 43). This step will involve deep analysis and synthesis of the gathered data to pinpoint areas where users have trouble organizing or being a part of events.

3.2.1 Creating of User Journey Map

A user journey map is a powerful tool for visualizing the user experience at different stages of interaction with a product. As Kalina and Nouri (2022) note in their work *Advanced UX Methods*: “A user journey map allows designers to see the product through the eyes of the user, identifying points of frustration and opportunities for improvement” (p. 201).

For creating of User Journey Map for application “Setkávač”, following steps were done:

- Definition of target group of potential users;
- Determination of actions, thoughts and emotions of the users;
- Identifying pain points and opportunities for improvement;

User journey map for event creation:

Stage 1 – Initiation

- Actions: The user decides to create an event and opens the application.
- Thoughts: “I want to arrange a meeting, but I'm not sure if it will be easy.”
- Emotions: Enthusiasm mixed with uncertainty.
- Pain Points: Fear of the complex process of creating an event.

Stage 2 - Creation of basic information about the event

- Actions: The user enters the name, date, time and place of the event.
- Thoughts: "Is this information enough? What else needs to be added?"
- Emotions: Concentration, slight anxiety.
- Pain Points: Uncertainty about necessary number of details.

Stage 3 - Privacy and invitation settings

- Actions: The user determines who can see and join the event.
- Thoughts: "I want new people to come, but I don't want too many strangers."
- Emotions: Uncertainty, desire for control.
- Pain Points: The difficulty of balancing between openness and privacy.

Stage 4 - Adding details and description

- Actions: The user adds a description of the event, tags, photos.
- Thoughts: "How to make my event attractive to others?"
- Emotions: Creative enthusiasm, slight stress.
- Pain Points: Uncertainty about which information is the most important.

Stage 5 - Review and publish

- Actions: The user views the created event and publishes it.
- Thoughts: "Did I do everything right? Will people come to my event?"

- Emotions: Excitement, anticipation.
- Pain Points: Fear that the details of the event might change (time, place).

Stage 6 - Post publication

- Actions: The user is waiting for the reaction of other users, possibly spreading the event in other networks.
- Thoughts: "How will I know if my event is successful? What should I do next?"
- Emotions: Impatience, slight anxiety.
- Pain Points: Uncertainty about the success of the event.

User journey map for searching and connecting to the event:

Stage 1 – Awareness

- Actions: User wants to find an event and goes to event's page.
- Thoughts: "I want to find an interesting event near me." "What events are happening soon?"
- Emotions: Curious, expectant.
- Pain Points: Process of finding an event might be complicated.

Stage 2 – Event search

- Actions: User finds relevant events based on preferences.
- Thoughts: "Do these events match my interests?"
- Emotions: Engaged, focused, potentially overwhelmed if results are too broad or too narrow.
- Pain Points: Overwhelming number of event options.

Stage 3 – Event evaluation

- Actions: User views the event details: photos, name, description, location, date, price.
- Thoughts: "Does this event fit my schedule?" "Is the price worth it?"
- Emotions: Interested, cautious, excited.
- Pain Points: Not having all necessary information about the event.

Stage 4 – Decision-making

- Actions: User clicks the "Attend" button.
- Thoughts: "Am I sure I want to commit to this event?" "What if something better comes up?"
- Emotions: Possibly hesitant.
- Pain Points: Not be able to change the decision.

Stage 5 – Pre-Event Engagement

- Actions: User participates in pre-event discussions.

- Thoughts: "I hope this event is as good as it seems."
- Emotions: Anticipation, excitement.
- Pain Points: Not be able to discuss upcoming event.

This user journey map reveals the key points where "Setkávač" can offer unique value and improve the user experience. For example, Li and Chen (2023) in their study "User Empowerment in Event Organization Platforms" note: "Platforms that give users a sense of control and confidence at every stage of event creation demonstrate significantly higher rates of user retention and event success" (p. 178).

3.2.2 Creation of "How might we?" questions

Based on the pain points that were mentioned in user journey maps (chapter 3.2.1) and using "How might we?" framework, certain questions were generated that in further stages can help us to improve the application.

Event creation:

- How might we simplify the process of creating events, making it accessible even to unexperienced users?
- How might we help users with filling up all necessary information about the event?
- How might we help the user to control the privacy of the event?
- How might we help the user to correctly add additional information about the event?
- How might we help the user with some uncertainty regarding the event?
- How might we provide the user with some tools for monitoring the event popularity and communication with potential attendees?

Searching and connecting to the event:

- How might we simplify the process of searching for the event?
- How might we decrease the number of the results that user will get?
- How might we make it clear what event is about?
- How might we help the user to change the decision?
- How might we help the user to discuss upcoming event?

3.3 Ideate

3.3.1 "How might we?" solutions

Based on the "How might we?" question above (chapter 3.2.2), in Ideate phase we are trying to find effective solutions to these problems, which will improve the application.

Mapping the questions with potential solutions:

Event creation

- Question 1: How might we simplify the process of creating events, making it accessible even to unexperienced users?
Solution: Provide an easy and intuitive entry into the event creation process.
- Question 2: How might we help users with filling up all necessary information about the event?
Solution: Provide hints and examples for each field.
- Question 3: How might we help the user to control the privacy of the event?
Solution: Offer flexible options for privacy settings. Switching between private and public event.
- Question 4: How might we help the user to correctly add additional information about the event?
Solution: Provide templates and tips for creating an engaging event description.
- Question 5: How might we help the user with some uncertainty regarding the event?
Solution: Provide a preview function and the ability to easily edit details.
- Question 6: How might we provide the user with some tools for monitoring the event popularity and communication with potential attendees?
Solution: Add a functionality to see how many people are going to attend the event, as well as chat functionality.

Searching and connecting to the event

- Question 1: How might we simplify the process of searching for the event?
Solution: Provide an easy and intuitive entry into the event search and connection process.
- Question 2: How might we decrease the amount of the results that user will get?
Solution: Add filter functionality while searching for the events.
- Question 3: How might we make it clear what event is about?
Solution: Display the event description so that the user can read more about the event.
- Question 4: How might we help the user to change the decision?
Solution: Provide the functionality to cancel the attendance.
- Question 5: How might we help the user to discuss upcoming event?
Solution: Add functionality to chat with other participants.

3.3.2 Paper prototype interpretation

Paper prototypes include sketches of the hand-drawn interface of the application. Using the data we received after conducting user research and defining the functionality of the application being developed, it is now possible to interpret that into a visual interpretation, i.e., user interface.

To find the optimal option and widen the scope for creativity, the crazy 8's method was used. Crazy 8's method

3.4 Prototype of Setkávač

Table 2 - Navigation routes

Exit point	Navigation routes	Purpose
Home screen	Home screen → Event screen	View of event details and connecting to it
Home screen	Home screen → Edit event screen	Adding new event
Profile screen	Profile screen → Editing the profile screen	Change personal data or customise
Event screen	Event screen → Event chat	Add messages to chat
Event screen	Event screen → Edit event screen	Event details update
Sign up screen	Sign up screen → Edit profile page	Set up/edit personal profile

3.4.1 Rules used in prototyping

The author used the rules of heuristics from chapter 2.4 and Gestalt principles from chapter 1.2.2 to design the application prototype. The relevant Gestalt principles and the rules of heuristics are listed for each screen in chapter 3.4.3 of the document.

3.4.2 Components

On the Figure below you can see components that were created in Figma (Figure 16), which include the *main visual elements* of the program.

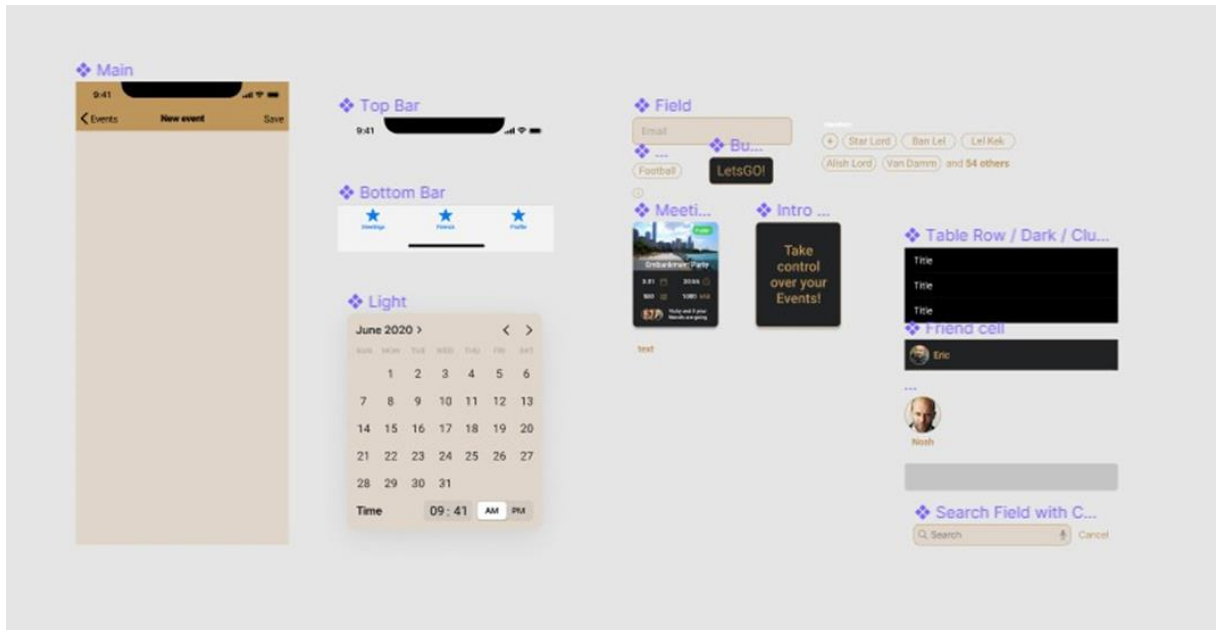


Figure 23 23 - Figma project

3.4.3 Functionality of prototype

On the figure below you can see how the starting page looks like.

Heuristics principles that were used:

- 8 - Aesthetic and Minimalist Design (welcoming screen with one button to continue within the application)

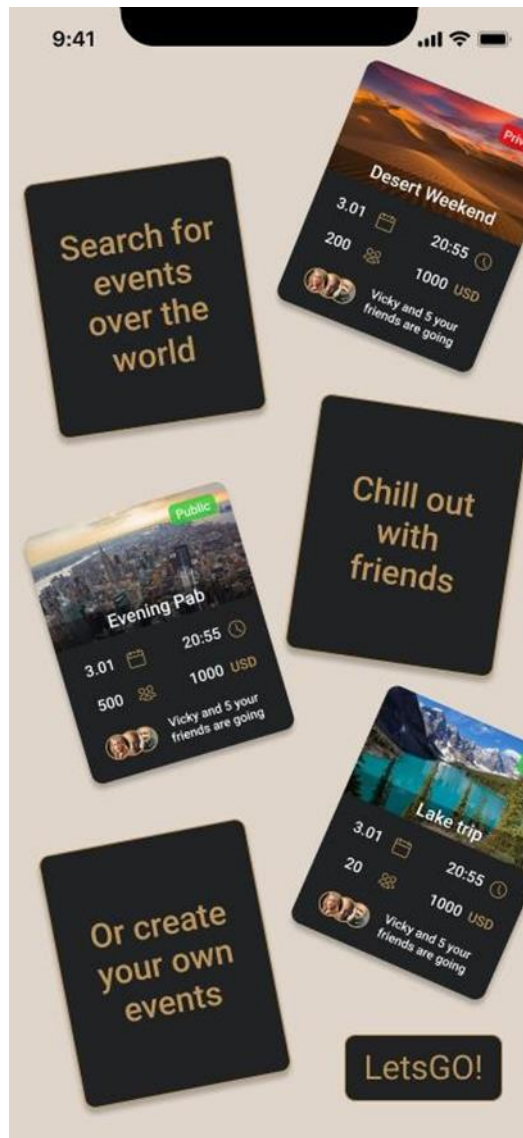


Figure 24 24 - starting page

Next, the user can click on the "LetsGO" button located in the lower right corner. After clicking, user is navigated to the next screen - Login (Figure 25)

Heuristics principles that were used:

- 4 - Consistency and Standards (users are used to this layout of login page)
- 8 - Aesthetic and Minimalist Design (no extra unimportant information)

Gestalt principles that were used:

- Proximity - input fields are closer to each other – sign in option. Further down is another option - sign up.
- Similarity – email and password fields have the same design
- Common region – email and password form a separate group – login form

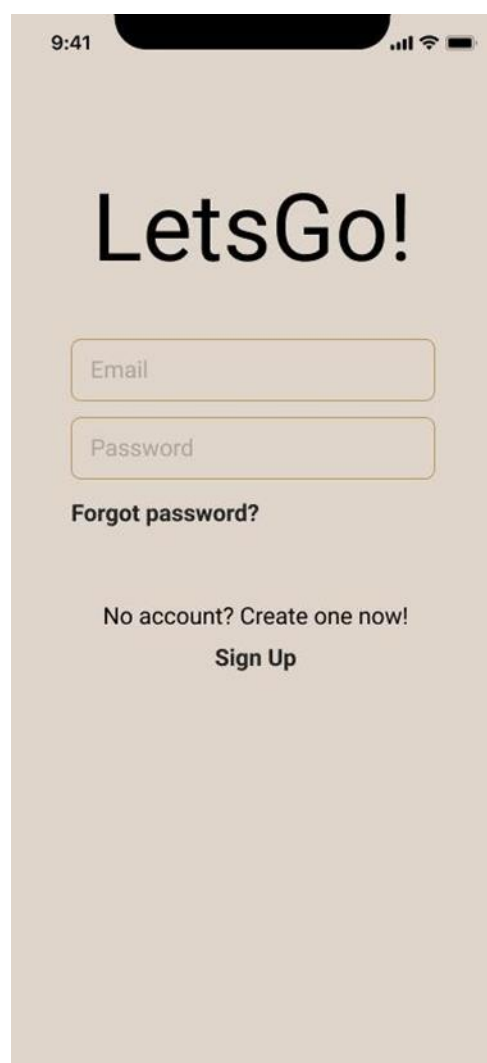


Figure 25 25 - login page

The functionality of the program covers the situation when the user forgets the password, so after clicking on the "Forgot password" button, he will receive a detailed instruction by mail. If the user does not have his own account, he can register by clicking on the SignUp button. The registration screen will appear (Figure 26).

Heuristics principles that were used:

- 4 - Consistency and Standards (users are used to this layout of sign-up page)
- 8 - Aesthetic and Minimalist Design (user just need to fill email and to create a password – no unnecessary fields)

Gestalt principles that were used:

- Proximity - input fields are closer to each other. Further down is another option - sign in.
- Similarity – email and password and confirm password fields have the same design
- Common region – email and password and confirm password form a separate group – sign up form

The image shows a mobile application interface for a sign-up page. At the top, the status bar displays the time 9:41, signal strength, Wi-Fi, and battery icons. The app's title 'LetsGo!' is centered in a large, bold, black font. Below the title, there are three vertically stacked input fields with rounded corners and a light beige background. The first field is labeled 'Email', the second 'Password', and the third 'Confirm Password'. All three fields have a thin orange border. Below the input fields, the text 'Alredy have account?' is centered, followed by a bold 'Sing in' button. The entire page has a light beige background.

Figure 26 26 - Sign up page

After entering the data, the application will show the login screen again with the email address prefilled. The user must confirm his email address before he can log in to his account.

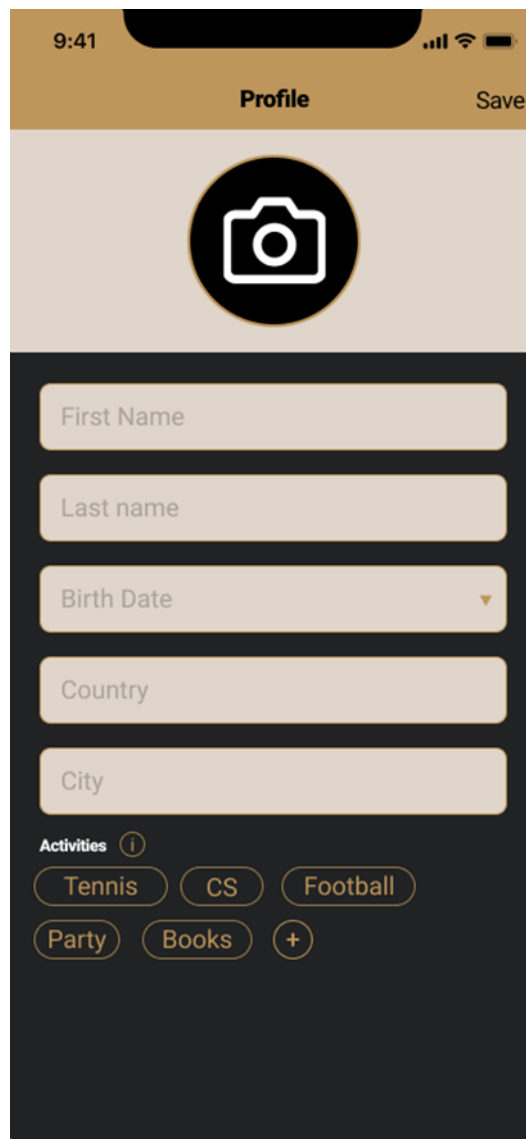


Figure 27 27 - Editing the profile page

Next, the user needs to fill out his profile (Figure 27). Each field will be analyzed and used to select the best event.

Heuristics principles that were used:

- 2 - Match Between the System and the Real World (understandable text explanation of every field)
- 4 - Consistency and Standards (starting with the main info about the user followed by additional info)
- 8 - Aesthetic and Minimalist Design (no extra unimportant buttons or text fields)

Gestalt principles that were used:

- Proximity - input fields are closer to each other, comparing to profile picture option or activities choosing.
- Similarity – all input fields have the same design, comparing to activities picker.

To select a profile photo, you need to click on the button with the icon “Camera”, which in will open the Photo Picker screen (Figure 28), that controls the system interfaces for taking photos, recording videos, and selecting items from the user's media library.

Heuristics principles that were used:

- 8 - Aesthetic and Minimalist Design (user just focuses on choosing the photo)

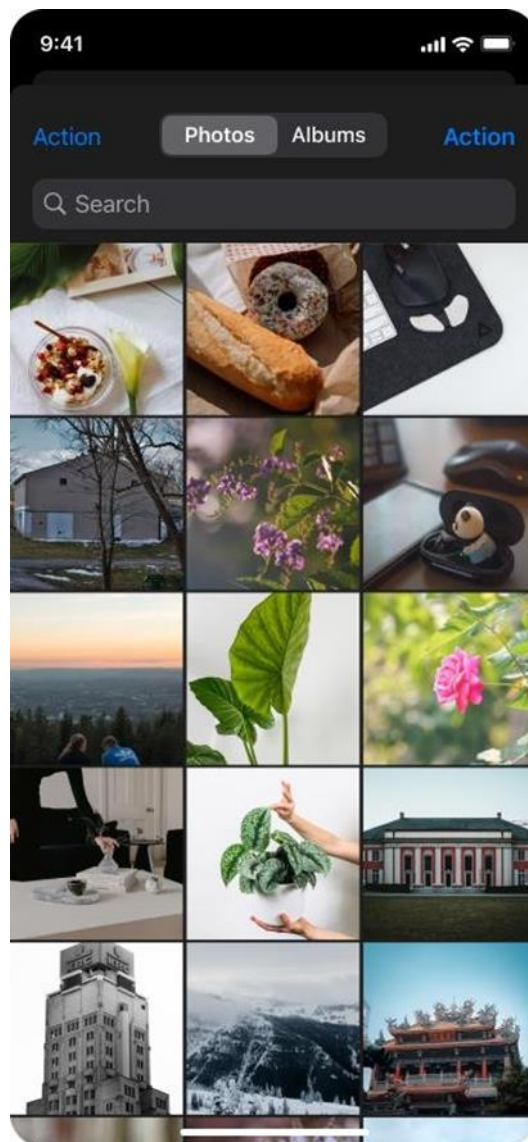


Figure 28 28 - Photo picker

There is a function to add so-called "Activities" on the profile filling screen. Thanks to them, it will be much easier to choose an event that will match the user's requests. After clicking on the "Info" button, a message will appear (Figure 29), which will describe why the function is needed.

Heuristics principles that were used:

- 10 - Help and Documentation (by clicking on info button, user can read the popup with info)

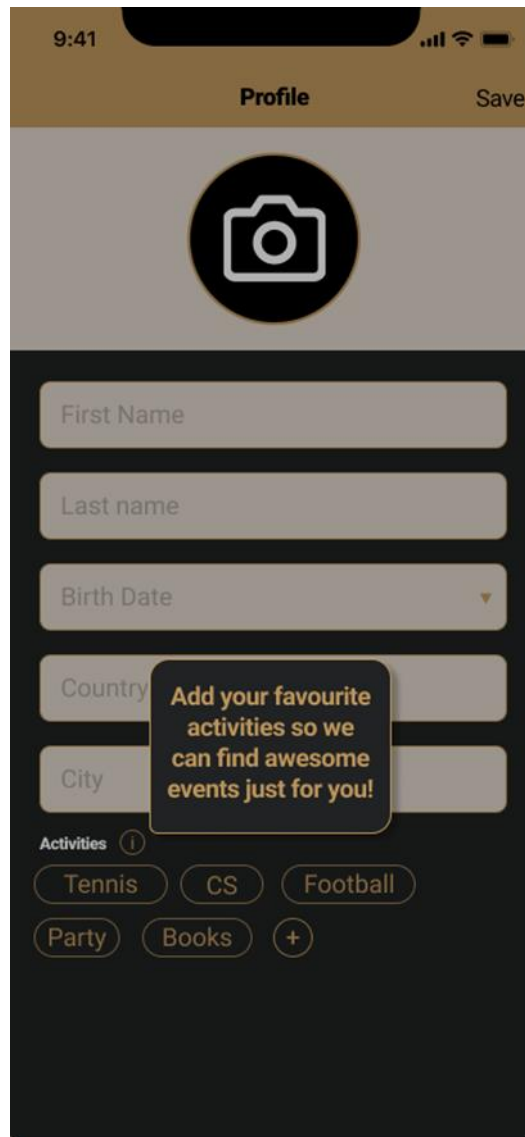


Figure 29 29 - Info popup

After pressing the "+" button, the keyboard with the helping panel will open (Figure 30). When the user types characters, it will show options that will be as much as close to the text that user types.

Heuristics principles that were used:

- 5 - Error Prevention (user gets a suggestion while typing)

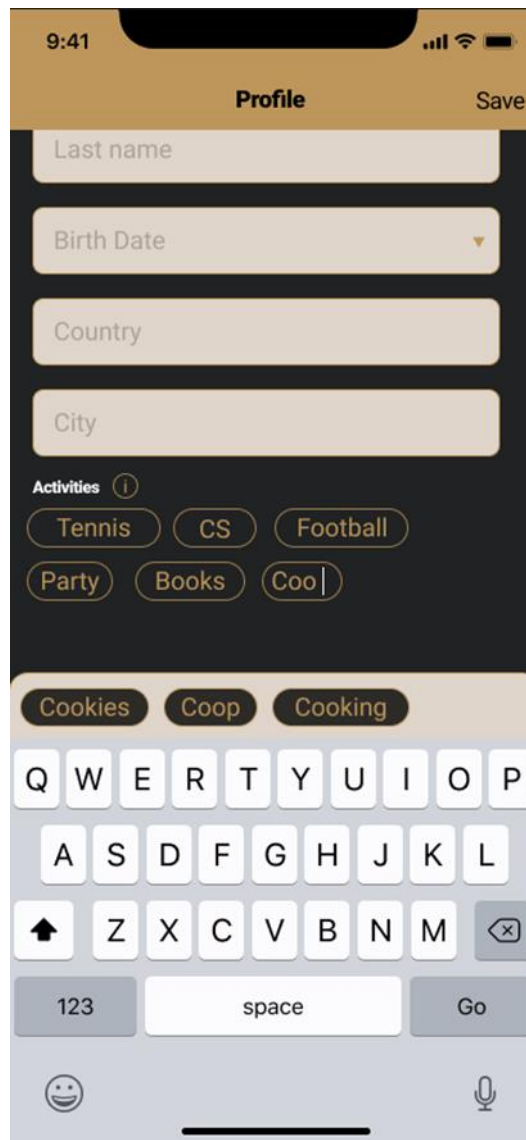


Figure 30 30 - Category suggestion

You can finish filling out the profile by clicking on the "Save" button in the upper right corner.

After that, a user with meeting records will appear on the screen (Figure 31). Meetings can be sorted (by date, activities, location, etc.) by clicking the appropriate button in the navigation bar on the right.

Heuristics principles that were used:

- 4 - Consistency and Standards (standard layout with top buttons with actions and list of events below it)
- 6 - Recognition Rather than Recall (buttons at the top are easy to recognize)

Gestalt principles that were used:

- Proximity – action buttons are closer to each other.
- Similarity – all events have the same pattern so user can easily identify them.
- Continuity – action buttons are placed in one line at the top right corner.

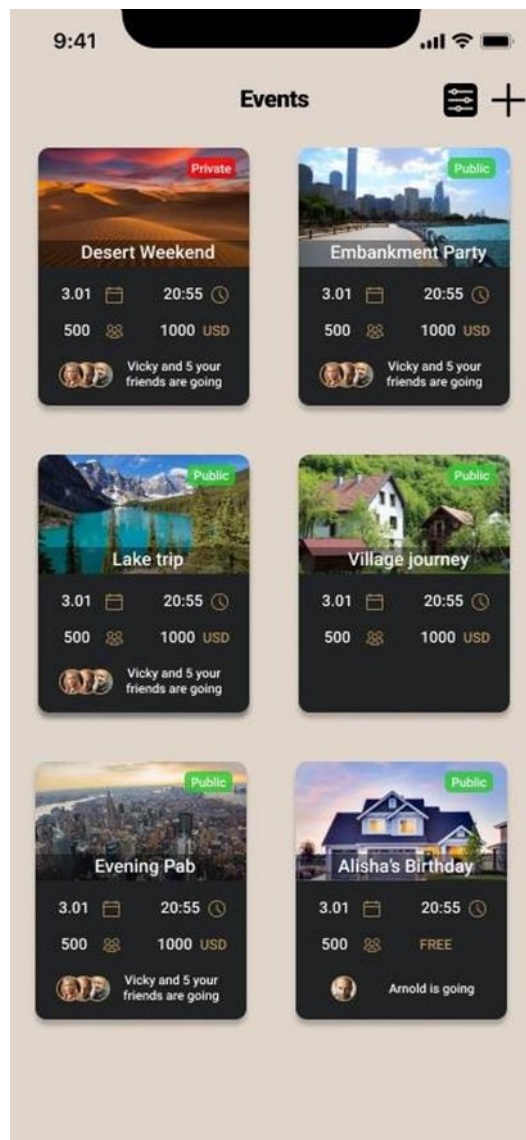


Figure 31 31 - Event's page

The list of meetings will be selected individually for each user. Both global meetings (for example a fan meeting with your favorite winemaker) and local ones (birthday celebrations, picnics) will be displayed.

The "+" button on the navigation panel will take the user to the meeting creation page (Figure 32).

Heuristics principles that were used:

- 2 - Match Between the System and the Real World (understandable text explanation of every field)
- 4 - Consistency and Standards (starting with the main info about the event followed by additional info)
- 8 - Aesthetic and Minimalist Design (no extra unimportant buttons or text fields)
- 10 - Help and Documentation (some fields have a placeholder to show the user in which format the data should be)

Gestalt principles that were used:

- Proximity – input fields are closer to each other
- Similarity – all input fields and options buttons have the same design.
- Continuity – members suggestions are placed in one row.

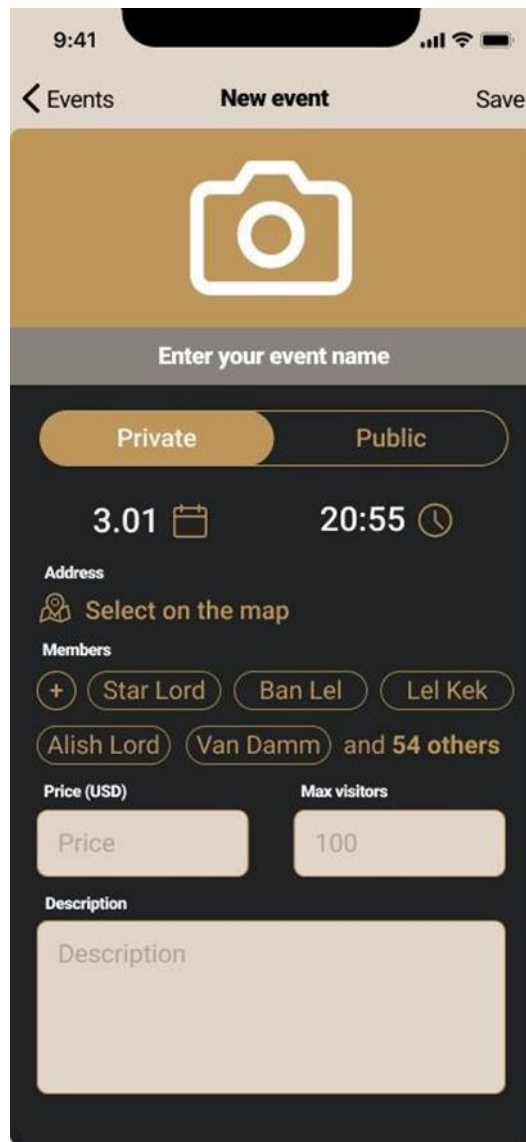


Figure 32 32 - Creation of the event page

The meeting photo is selected the same way as the profile photo.

Meetings can be both private and public. You can join the public ones at any time, if you have not yet reached the maximum number of participants.

The date and time of the meeting are selected using the Date Picker (Figure 33)

Heuristics principles that were used:

- 4 - Consistency and Standards (very understandable layout of choosing the date as in most of other applications)

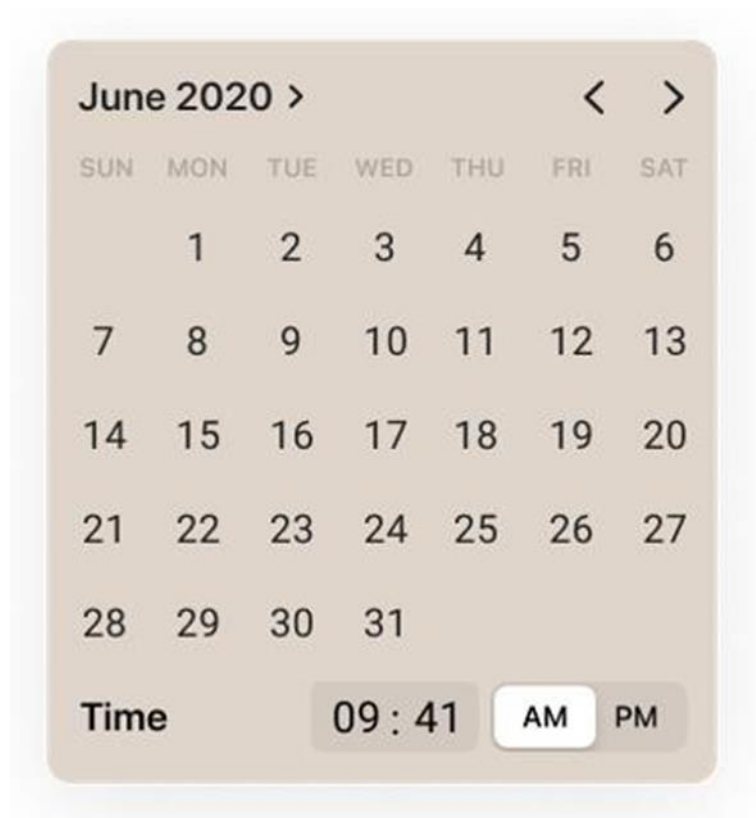


Figure 33 33 – Date Picker

It is possible to immediately invite friends according to the principle similar to the Activity. The price of the visit is indicated, or is left blank if the event is free. The number of invitees is also indicated only if necessary.

You can leave any additional information in the description: dress code, page in social networks, rules of conduct, etc.

To choose a meeting place, you need to click on the "Select" button on the map, after which the screen for selecting the location on the map will be displayed (Figure 34).

Heuristics principles that were used:

- 1 - Visibility of System Status (user automatically sees the “GO” pointer at his current location)

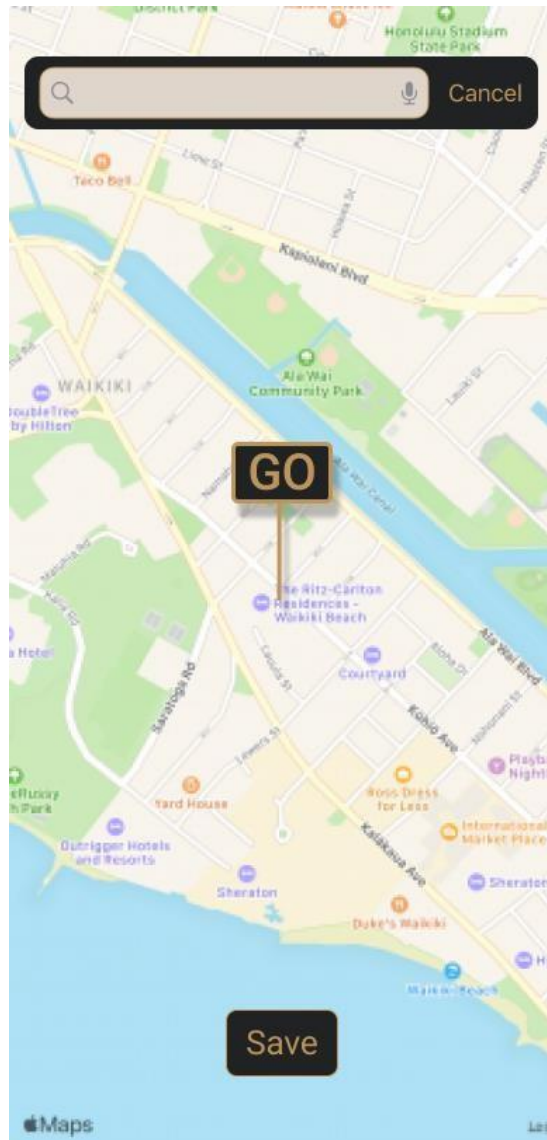


Figure 34 34 - Map for choosing the location

The table with the label "GO" is an annotation, the coordinates of which will be translated into a real address and saved on the server. There is also an option to find a location using the search bar at the top. The "Save" button will return the user to creating the meeting.

The next step is to open the friends screen (Figure 35). In the future, you will be able to create meetings and share them with friends.

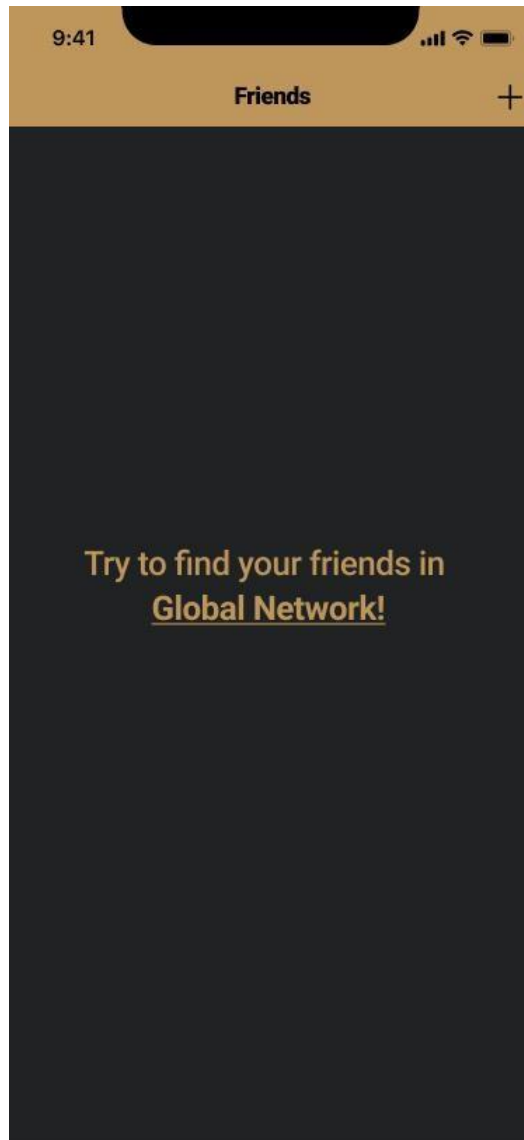


Figure 3535 - Friends display

If the user has no friends inside the application, the label "Try to find your friends in Global Network!" will be displayed. Clicking on the highlighted label will open the global network screen with all users (Figure 36).

Heuristics principles that were used:

- 2 - Match Between the System and the Real World (user easily understands that he/she needs to add friends by clicking on Global Network button)
- 8 - Aesthetic and Minimalist Design (no extra unnecessary buttons or texts)

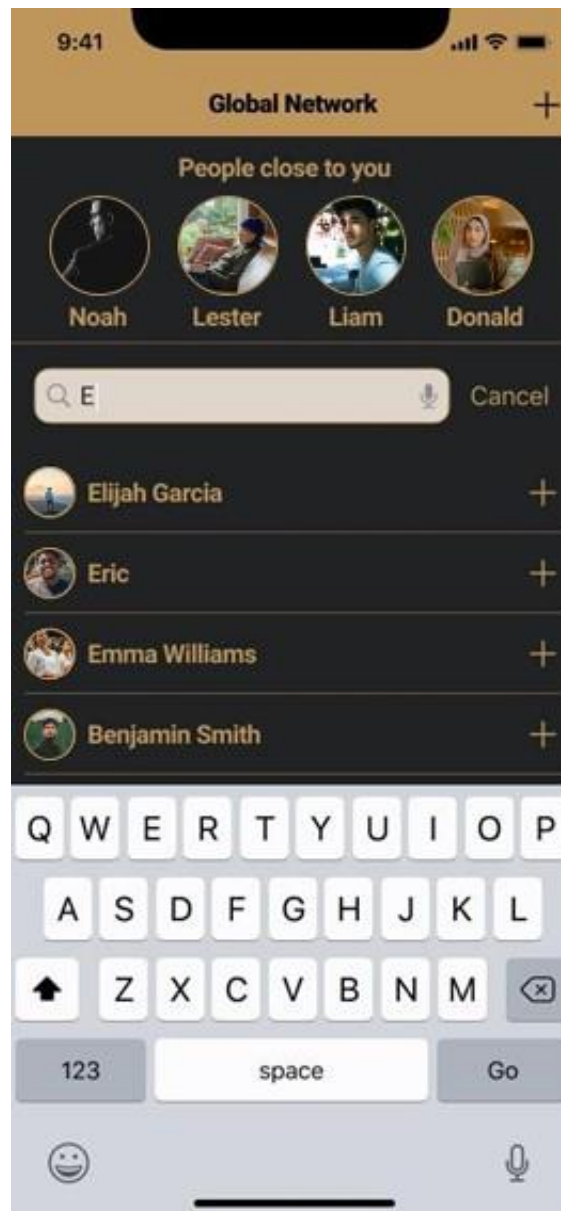


Figure 36 -36 Display with searching for friends

Here you can find both your friends and other users around the world. A separate function can be used to define the possibility of adding people nearby as friends. The panel will be hidden if the user denies access to their location.

Heuristics principles that were used:

- 5 - Error Prevention (while typing the name, user gets the suggestions of potential people he/she is looking for)

Gestalt principles that were used:

- Common region – people that are close to the user are displayed close to each other horizontally are separate from the people that the user is searching for, who are displayed vertically.
- Similarity – users' profiles have the same design style.

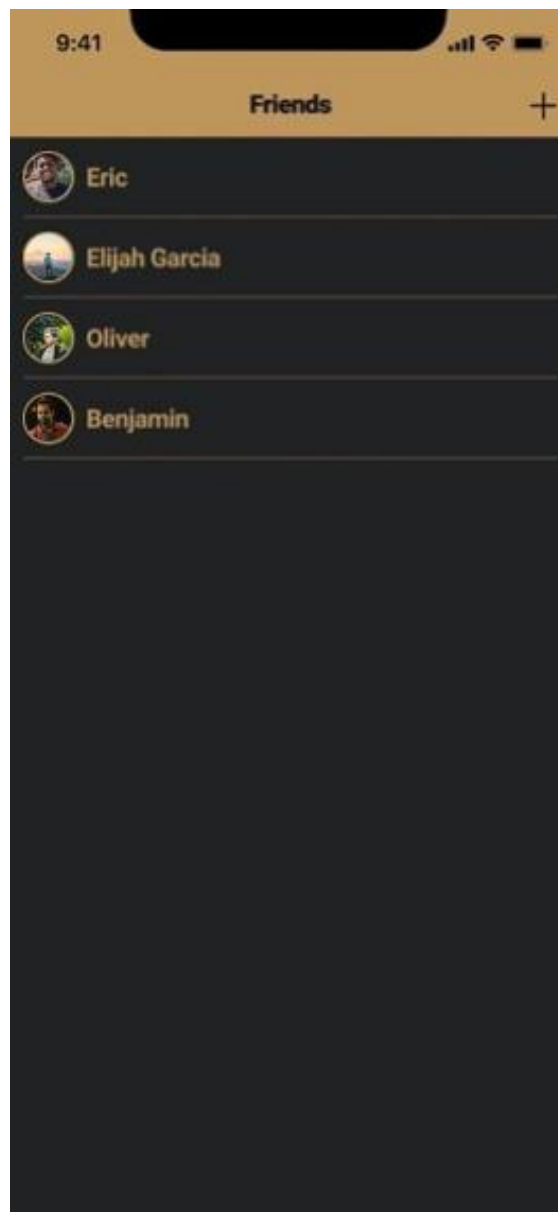


Figure 37 37 - Filled display with friends

Also, each meeting will have its own chat where users discuss everything regarding the current event (Figure 38).

Heuristics principles that were used:

- 2 - Match Between the System and the Real World (user can easily understand the functionality of the buttons by their icons)
- 3 - User Control and Freedom (user can easily quit the chat by pressing “event” button and return back)

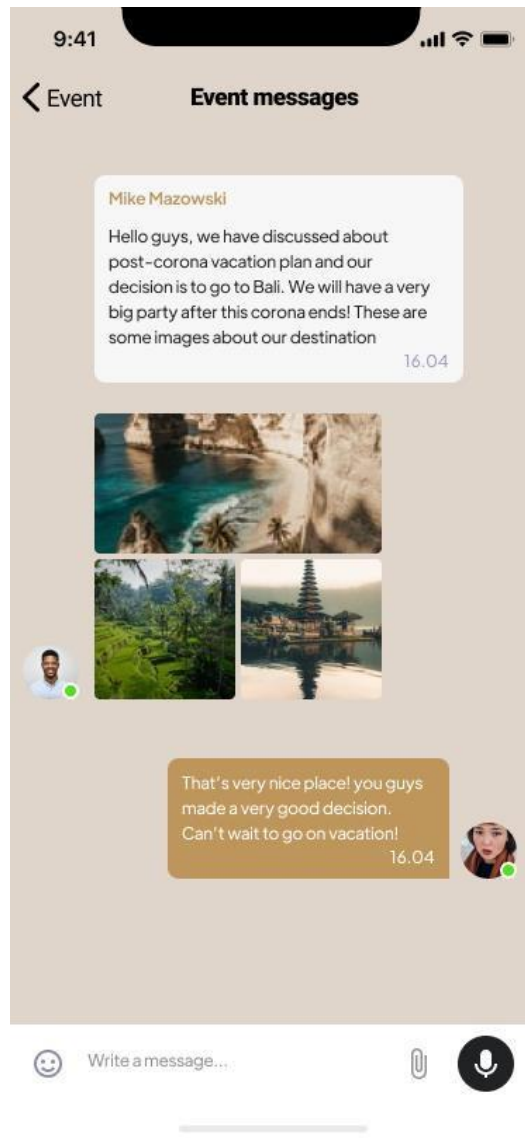


Figure 38 38- Chat

With the help of control element that displays one or more buttons in the tab bar for choosing between different subtasks, types or modes of the program, user will be able to switch between the main screens.

Heuristics principles that were used:

- 4 - Consistency and Standards (tab bar at the bottom is a classical solution for most mobile applications and easy to understand)

Gestalt principles that were used:

- Continuity – all tab menu options are displayed in one line at the bottom of the application

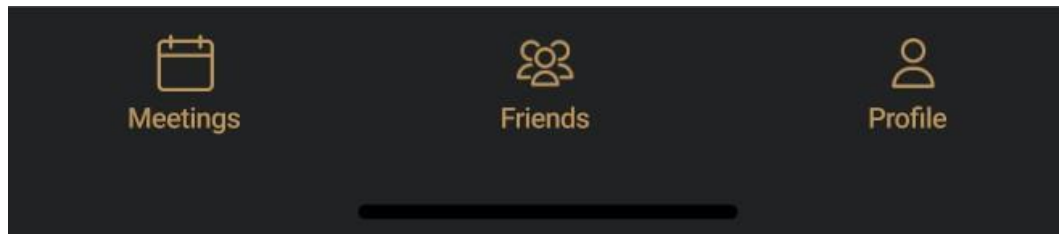


Figure 39 39 - Tab bar

3.5 Testing

Usability testing is necessary

3.5.1 General approach

Based on the prototype that was created before, next step was to test the usability of this prototype. Then all the necessary steps to handle the test were defined:

- Have a prepared prototype for use
- Identify the task that participants will have to complete
- Recruit the participants
- Choose the most comfortable testing method
- Conduct the test
- Analyse the results

3.5.2 Execution of the testing

Identifying the tasks

To handle the testing, certain test scenarios were created. They were covering the main functionalities of the application; list of the tasks is written below:

- Sign up in the application and set up the profile
- Create a new event
- Search for a certain event and join it

Recruit the participants

The participants consisted of event organizers, event attendees, and casual users who are interested in these activities. Altogether there were 6 participants that fell under the determined target audience characteristics in chapter 3.1.2

Table with participants is demonstrated below:

Table 3 - List of the participants

	Sex	Age	Technical Knowledge
Participant 1	M	18	High
Participant 2	M	21	Medium
Participant 3	M	32	High
Participant 4	F	23	Medium
Participant 5	F	42	Low
Participant 6	F	50	Medium

Choosing of the method

For our testing goals, In-person method for testing has been chosen. In-person usability tests take place offline when users are asked to complete a variety of tasks in the presence of a researcher or alone (Radek, 2024).

Conduction of the tests

One-to-one in-person meeting was held with every participant. At the very beginning of the testing, participant was informed what they are going to do and the purpose of testing.

Every user got 3 scenarios that they had to complete. After completing the scenarios, they were asked for additional questions:

- How did you feel about the navigation of this app?
- What was the creation event process easy and intuitive for you?
- What was the joining the event easy and intuitive for you?
- What did you like about the app?
- What did you dislike about the app?

3.5.3 Results

Analysing the results

Based on the testing result, all the participants (100%) where able to fulfil the scenarios that they had to do. Also, it has taken around 1 minute 3 seconds on average to sing up in the application and set up the profile, 1 minute 13 seconds to create a new event and around 39 seconds to find a certain event.

Also participants marked good and intuitive design of the prototype. Respondents correctly recognized the parts of the application that belonged together which testifies the correct use of Gestalt and Heuristic principles.

Test limitations

There are certain limitations that can decrease the results:

- Only 6 participants have attended the testing, for better results it would be better if more people have participated.
- Figma functionality restrictions – as it is not a real application, testing was limited.

3.6 Place for improvement

Now, the main functionality of the application has been created, but it is still under development. It is planned to add such features as:

- Displaying users on the map
- Expanding the creation of meetings with lists of tasks
- Integration with social networks
- Possibility of individual correspondence

Conclusions

The objective of this bachelor's thesis was to create a mobile application interface with the highest level of use, convenience, and aesthetic appeal for a user.

The process of implementing a mobile application design that has been carried out with the help of design thinking practices. The following tasks were completed during the process:

- Defining mobile applications and choosing the best way for our application
- Defining UX methodologies and choosing one that suits us the best
- Going through the whole Design Thinking process in order to create a prototype
- Test and evaluate the prototype

In contrast, the completed design appears clean. Future plans on this app include individual correspondence, integration with social networks, displaying users on the map and expanding the creation of the event with the list of tasks. After completing the task, we can state that design that was created is ready to be integrated to the application in order to test it real situations.

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Appendices

Appendix A: Semi-structured interview scenario

Start of the interview

Hello, I'm glad you decided to take part in the testing. My name is Elina Salnikova and I will carry out our entire interview.

First, I will talk about the application. I'm trying to design a mobile application for people to network and connect by creating or finding events and meet ups near you. The purpose of this interview is learning your experience with similar applications and what you would like to see in the new one.

Every piece of information from our interview is valued so, please, feel comfortable with giving honest feedback. The gathered information will only be used for research purposes.

Our interview will last about 30 to 50 minutes. Do you have any questions?

Do you agree with recording the interview?

Main part of the interview

What experience do you have regarding social events?

How often do you take part in social/networking events?

Have you ever organised/hosted events?

Have your socialisation activities rate change over the course of last several years? Why?

How was your experience with applications of similar type?

How was your experience with event discovery on these apps?

Have you created any events? If yes, how was your experience with it?

What makes you frustrated of these apps and what do you think could be easier/better?

Which features would you like to see in a perfect application for event organisation and discovery?

End of the interview

Is there anything else that you want to talk about, that we didn't cover?

Thank you for your time. You have provided valuable data that will contribute in development of a design that is simple to use and satisfies its users' needs.

Appendix B: List of tasks and research questions within the usability testing

The participants were asked to complete the following tasks

1. Launch the app and go through sign up process

- a) Did you have any problems completing this task?
- b) Do you have any ideas for improving this process?

2. Try creating a new event

- a) Did you have any problems completing this task?
- b) Do you have any ideas for improving this process?

3. Try joining an event

- a) Did you have any problems completing this task?
- b) Do you have any ideas for improving this process?

After carrying out the tasks, the participants were asked the following questions

- How did you feel about the navigation of this app?
- What was the creation event process easy and intuitive for you?
- What was the joining the event easy and intuitive for you?
- What did you like about the app? And why?
- What did you dislike about the app? And why?